Knowledge-based Extraction for Inverse Design of Optical Thin-Films

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***Abstract*—In this paper, we've developed an inverse design framework for optical thin-films assisted by Generative Pre-trained Transformers (GPT). As a Large Language Models (LLMs)-powered tool, this creates an integrated platform for users to perform various functions related to inverse design on thin-films using an accessible and unique natural language interface via the assistant tool. With the consultant tool, users can query important design information and receive relevant answers to enhance the design knowledge of optical thin-films. This platform proves capable of speeding up design of thin-film structures compared to traditional ways and can also save many hours for repeated knowledge querying. The developed platform significantly reduces the resources and time (~3x) for designers in developing optical thin-films to reach an optimal design. This will allow the thin-films' applications to be brought to the market faster and with increased efficiency.**

Keywords—optical thin-films, transfer matrix method, inverse design, knowledge-based extraction, large language models

I. Introduction

Since the past few decades, multilayer optical thin-films have been extensively developed to be used in an extremely versatile range of applications. These thin-films consist of layers of various materials with thicknesses that can be carefully controlled. This allows designers to precisely adjust thin-films' optical properties to suit a variety of use cases. An application of thin-films is in radiative cooling, where thin-films allow passive dissipation of heat and minimal absorption of ultraviolet and infrared spectra wavelengths to avoid unwanted warming for energy-saving purposes. They can be applied in photovoltaic cells, making these cells substantially more cost-efficient. They even have critical applications in biomedical imaging, enabling clearer pictures such as in colonoscopies where certain features are accentuated. The diverse applications of thin-films have rendered them an integral part of society. However, the thin-film design process can be time-consuming.

For designers with limited theoretical background or design knowledge, it can take weeks to yield a satisfactory design. It is also time and cost-inefficient to repeatedly redesign similar device functionalities for each application niche [1-4].

Predicting the behaviour of light (i.e., transmission, reflection, and absorption) through multilayer optical thin-films can be solved using modelling techniques such as the Transfer Matrix Method (TMM). However, optimising the structure of thin-films to achieve the desired performance can be a challenging task due to the large number of parameter combinations and the non-linear relationships between the parameters and the optical properties of the thin-film. Nevertheless, there currently exist various state-of-the-art methods of efficiently designing thin-films. Inverse design is a popular design method that produces an optimal thin-film structure based on the designer's given performance or metrics requirements. Algorithms used for thin-films inverse design include local gradient-based methods (back propagation), global optimisation such as evolutionary algorithms, and Deep Reinforcement Learning (DRL). However, designers with limited knowledge or experience with design methodologies may find choosing and implementing suitable inverse design algorithms challenging and time-consuming (e.g., trial and error), further slowing down the development of thin-films' applications [5-8].

Natural language processing in Large Language Models (LLMs) such as in OpenAI's GPT-3.5 pre-trained conversational model has proven capable of greatly speeding up development in specific domains. Natural language interfaces enhance the usability of tools, with the potential to augment designers' knowledge about relevant methodologies to accelerate the optical thin-film inverse design process. Having been trained on large textual datasets, LLMs' natural fluency and ability to converse well with people allows them to act as a middleman to streamline optical thin-films development.

While LLMs have shown significant improvements in their ability to comprehend natural language, their generic training datasets make them too generalised for specific use cases. Moreover, they may be prone to hallucination and generating false information. With the implementation of specific instructions and contexts for LLMs (i.e., fine-tuning), it can then become possible to achieve the desired output from a simple description of the designers' task. LLMs are also an important source of design knowledge and information in formulating responses in an easily understandable manner [9-11]. This paper presents an implementation of an LLMs-powered agent to make a versatile and accessible inverse design tool to aid engineers in thin-film design. The developed platform will be a multi-LLMs-based intelligent agent capable of performing inverse design (an assistant) and knowledge retrieval (a consultant). It can overcome the above-mentioned challenges, solving the bottlenecks that prolong the time taken from design, optimisation and testing to prototyping of optical thin-films.

II. Architechture of the Developed Platform

The platform has been developed allowing designers to interact directly with the conversational LLMs. Its architecture consists of two separate features—the assistant is capable of understanding necessary parameters to perform inverse design and the consultant answers questions in an expert manner by querying relevant knowledge related to optical thin-films design, from theory to methods and applications. The consultant hence efficiently supports the usage of the assistant tool by providing targeted information like thin-film configuration and design techniques. The tool is able to plot relevant graphs and output the data for post-processing and analysis. These form an integrated portal which can design thin-films effectively. We utilise OpenAI's GPT-3.5 Application Programming Interface (API) which has already been primed with its context set through an initial system message. The GPT-3.5 model acts as the front-end for users' natural language input, which can then call the appropriate assistant or consultant functions. Fig. 1 shows the full workflow of the platform for both of its capabilities.

Relevant up-to-date documents (*.pdf, *.doc, *.ppt) curated on the topic of optical thin-films (arXiv, Google Scholar, and internal documents) are organised in a database and are parsed. These files are then split up into information chunks of equal length to generate embeddings from each chunk, which are stored in a vector store. When the consultant is used, OpenAI's Embeddings API creates embeddings from the user's question. By searching for similarities between the question's and information's embeddings, relevant chunks of information are selected. These pieces of information are then sent in a prompt to OpenAI's GPT-3.5 API to formulate the answer for the user, generating more accurate and specific responses to questions. Through the provision of relevant data, the tool has the ability to provide much more specific and contextual answers instead of generic responses.

The added expertise in its answers based on peer-reviewed and recent research provides users with guidelines in structuring their optical thin-films design approach. It also reduces the likelihood of falsifying information, instead focusing more on processing information from the documents into a clearly-worded answer.

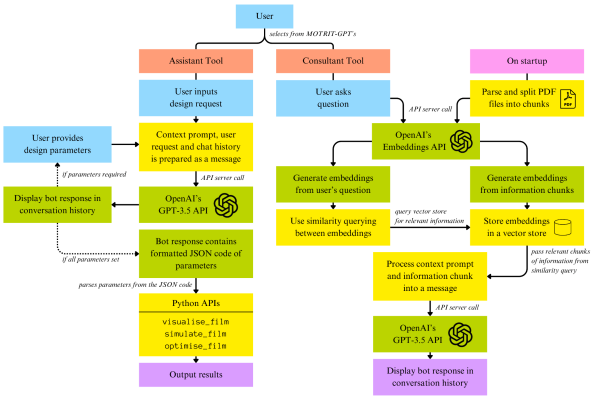

Fig. 1. Workflow process of interaction between users and the developed inverse design platform to produce results. (Left) Assistant workflow. (Right) Consultant workflow.

III. LLM-based Knowledge Extraction

The consultant capabilities focus on providing design knowledge and applications information regarding multilayer optical thin-films. Given a specific user question about multilayer optical thin-films, it summarises, synthesises, and provides information from the latest research and expert sources. This formulates an informative yet contextually correct and concise answer that addresses the question with peer-reviewed information and avoids making up data. This problem is particularly pertinent as inexperienced designers may need more insights in understanding the physics behind optical thin-films, their design configurations, processes, purposes and more. This can quickly provide them with a reliable output instead of having to spend long hours researching to answer their question. The ability of the consultant was proved by asking for information about TMM-Fast, a recently published method of simulating optical properties of multilayer thin-films that users may find relevant if they are exploring the ability to accelerate the simulation of thin-films. Fig. 2 shows a comparison of the consultant's response and the conversational GPT-3.5 model's response to the same question regarding TMM-Fast. It has been seen that the consultant can provide a substantially detailed response about TMM-Fast. Comparatively, GPT-3.5 can only indicate a "lack of knowledge" of this package. The information is also verifiable against the work of Luce et al. (2022) who developed TMM-Fast [5], proving that the likelihood of the consultant falsifying information is minimal. The demonstration proves that the consultant is versatile and is able to provide expert information, giving it an edge over generic LLMs. It is also faster compared to manually searching through large article databases, providing significant time savings, especially for repeated knowledge querying.

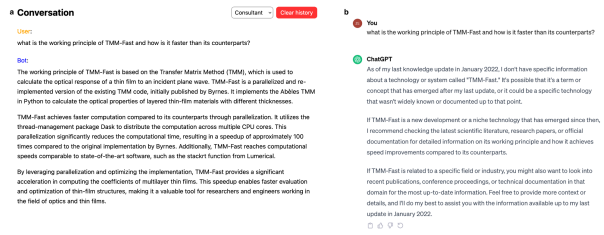

Fig. 2. Comparison between Responses regarding TMM-Fast. (a) The consultant's detailed and informative response about TMM-Fast. (b) OpenAI's pretrained conversational GPT-3.5 model's generic response (obtained from OpenAI's web interface at https://chat.openai.com/).

IV. Inverse Design of Optical Thin-films

A. *Knowledge Extraction-assisted Inverse Design*

In the assistant, the pre-trained GPT-3.5 model as an LLM leverages its memory of previous conversations and understanding of available functions and associated parameters to prompt users as needed for data. It repeatedly prompts the designers, guiding them to provide more information regarding the parameters until sufficient details are acquired. The LLMs then formats the parameters into JavaScript Object Notation (JSON) code and triggers specific messages to call server-side Python functions. These inverse design-related functions then automate calculations to generate the desired configurations and data.

The assistant can converse well in natural language. It prompts designers for the parameters as required and outputs them in the desired format, smoothly guiding them through the process while the assistant's responses adhere to given instructions. When sending an API request to OpenAI's conversational GPT-3.5 API, there is a need for the model to be informed of its role and function. This helps to avoid very generic responses from OpenAI's conversational model (Fig. 3). As shown in Fig. 3, the response provided by the conversational GPT-3.5 model is not that helpful for those who want to carry out autonomous design or functions on their optical thin-film for further investigation. As the developed platform uses a system prompt, it ensures that all necessary parameters are given and correctly formatted for parsing. The key elements of the system prompt are the tool's purpose, the Python API's functions, the required parameters, and their associated formatting. By utilising a system prompt, it ensures the model understands its instructions to interact with the user meaningfully. Overall, the assistant can reduce the time taken for the design process by up to 90% compared to manual design, significantly reducing time-intensity.

A typical inverse design function requires the number of materials (n), materials list (mat_list), bottom substrate (sub), the medium above thin-film (top_med), range of wavelengths of light (λ_0 to λ_1), and angle of incidence of incoming light (θ). As an example of the inverse design process, the design target function of a reflector is defined by specifying the desired spectrum through these parameters: the general reflectivity of the thin-film (R), the resonant wavelength or wavelength ranges (λ_r) and the resonant reflectivity (r). For narrow-band optimisation, λ_r are singular values, whereas for wide-band optimisation, λ_r are a range of values of resonant wavelengths. The spectrum will then have a reflectivity of R at all wavelengths except for λ_r, where it will have a resonant reflectivity of r. Wavelengths around λ_r will also not have reflectivity of R, instead having reflectivities gradually approaching r modelled by a Cauchy distribution. Fig. 4 (right) provides a depiction of the target spectrum. With the focus on reflectors for this problem subset, the desired thin-film is only optimised for reflectivity in this use-case.

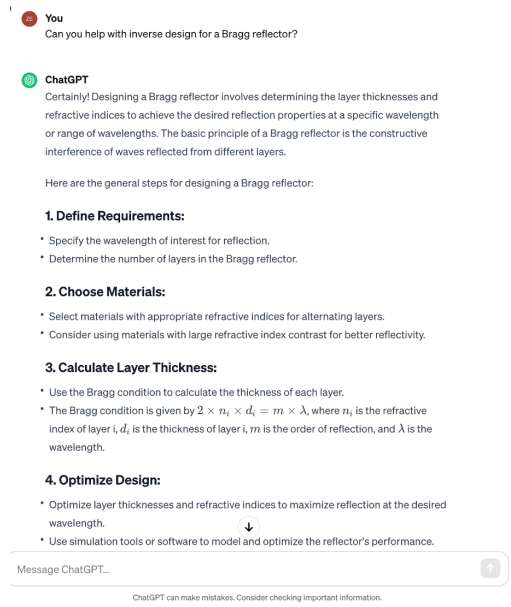

Fig. 3. Generic Response by OpenAI's Pretrained Conversational GPT-3.5 Model Without Prior Context Setting (obtained from OpenAI's web interface at https://chat.openai.com/). The generic response to an inverse design problem provides only a few general steps to design the reflector.

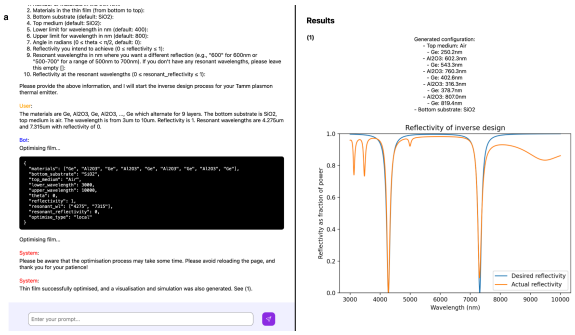

Fig. 4. Local gradient-based inverse design process of narrow-band Tamm plasmon thermal emitters at the long-wave infrared range with resonance for CO_2 and SO_2 non-dispersive infrared sensing (conversational history on the left has been truncated).

B. *Inverse Design Process*

We've implemented inverse design approaches tailoring to the needs of the state-of-the-art optical thin-films design taking into consideration the design knowledge and information output from both consultant and assistant tools: the local gradient-based and the global optimisation-based inverse design. In the local-based approach, while designers can control a larger number of parameters they possess for the design and the optimisation process is generally faster, it tends to converge to the local

minima. Regardless, the optimisation of narrow-band spectrum still yields extremely good results. Fig. 4 provides an example of the use of the local gradient-based inverse design for obtaining optimal configuration of a narrowband optical thin-film reflector [2]. As in Fig. 4, the local-based inverse design starts with all thin-film layers taking random thicknesses which are then iteratively adjusted. By backpropagation of the fitness value compared to the actual spectrum, a stochastic gradient descent-based optimisation process occurs to find a local minimum in the fitness function, generating the optimised configuration.

The global optimisation-based approach leverages on a hybrid technique [7]. It begins with differential evolutionary optimisation to select a set of thin-film designs that are close to the global minimum. This optimised set is used as the initial geometries for a gradient-based optimisation process through the neural adjoint method. This neural adjoint method is conducted using a Residual Network trained on thin-films and their emission spectra as the forward model. With this optimised initial population, the neural adjoint converges significantly faster at the global minimum for a more accurate design. Fig. 5 presents a use-case of broadband optical thin-films reflector (Bragg reflector), which is optimised by using global optimisation-based inverse design. For this use-case, the local gradient-based approach may provide an outcome which is far from the global minimum as inverse design of broadband devices using local optimisers can lead to poorer fitting results. On the other hand, in Fig. 5, the global-based optimiser can find a minimum closer to the global minimum to achieve a more optimal design. The current implementation of the global-based optimiser also does not require the parameters n and mat_list to be specified. This is as the implemented hybrid global-based inverse design is more generic, providing slightly less control over the thin-film parameters. As in Fig. 5, even though the desired spectrum is highly theoretical, the suggested results still demonstrate a close match with the correct resonant wavelengths showing high accuracy in the generated optimal configurations.

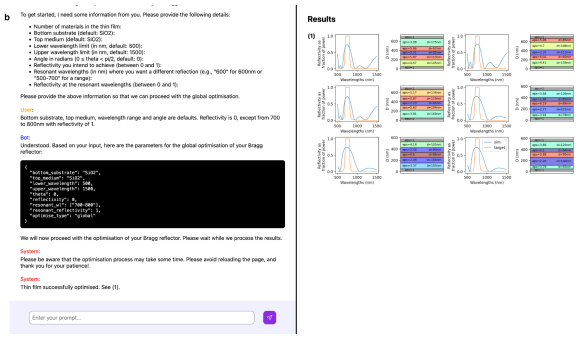

Fig. 5. Global optimisation-based inverse design process of a wide-band Bragg reflector from 500 nm to 1500 nm with resonance at 700–800 nm. "eps" (epsilon) indicates the permittivity, equivalent to the square of the refractive index of that layer (truncated conversational history on the left).

V. Conclusions

We've developed a novel inverse design framework, which pools all the resources and functionalities regarding multilayer optical thin-films into a single design platform. By using natural language as its input, it makes the user experience highly guided to develop optical thin-films with ease. The platform has proven capable of accelerating this process, cutting away hours of work as demonstrated in the above use cases, e.g., inverse design of Bragg reflectors, with both the consultant and assistant tools. Overall, thin-film engineers using this LLMs-powered tool can generate designs faster, test them meticulously, and bring them to the market efficiently with a higher probability of achieving an optimal design. With the widespread application of optical thin-films, it is no doubt that this will be able to further our understanding of thin-films in various domains. Future work includes creating even more integrated experiences. For example, we may combine the two tools into a single AI agent to avoid having to switch between the tools. This can even allow the consultant to suggest functions for the users to run to better understand their thin-films. We can also increase the flexibility and shareability. This can be done by producing downloadable reports of the results, customising diagrams, or exporting raw data instead of just graphical data. The assistant's functions can also be further enhanced by integrating DRL, which is a powerful method in performing inverse design of optical thin-films. DRL algorithms such as Deep-Q Learning and Proximal Policy Optimisation can be used to suggest not only thicknesses, but also realistic materials to achieve the intended spectrum. These methods can provide good starting points for designers instead of having to experiment with a plethora of possibilities. This will further enhance the ability of the developed platform to aid in the research of optical thin-films.

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